IMPERIAL

Department of Computing
70015: Mathematics for Machine Learning
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Practice Problems 01: Problem Settings & Linear Models

ML Settings

Note for students: These kinds of questions are unlike those that you will see in your final exam, but are meant to help you begin thinking beyond just the settings that we will explicitly cover.

Question 18 (Encoding Bird's Eye View LiDAR Representations (Opening to Alphabet Interview Question))**.** Suppose you are given a LiDAR scan of the environment around a car. The LiDAR data have already been projected into a *bird's-eye view* (BEV) representation. You are asked how you would formulate the feature space for this BEV under two different modeling scenarios:

a. A model that accepts *images* as inputs.

b. A model that accepts *text* as inputs.

Describe what the feature representations might look like in each case.

Question 19 (Transfer Learning)**.** Transfer learning is a machine learning problem setting where a model trained on a source dataset is used on a target (sometimes called down-stream) learning task that is different but related. In this setting:

1. Using set theory notation, formulate the source dataset $\mathcal{D}_{\text{source}}$ and the target dataset $\mathcal{D}_{\text{target}}$.
2. If we wish to use a linear model, what aspects of the datasets must be related? Explain your reasoning.

Question 20 (Continual Learning)**.** Continual learning addresses the problem of learning over multiple tasks. One can think of this like transfer learning except we are now going to train on data from both the source and target datasets. One can assume there are k different tasks. Consider the following settings:

1. **Domain-Incremental Learning (DIL)**: All tasks have the same label space $\mathcal{Y}$, but different input distributions and therefore different joint distributions. Use probability notation to formalize this.

2. **Task-Incremental Learning (TIL)**: Each task has a disjoint label space. Use set theory notation to formalize this. How might you still use the same model for these settings?

3. **Class-Incremental Learning (CIL)**: Each task has a disjoint label space. Task identities are only provided during training. Use any notation you would like.

4. **Online Continual Learning (OCL)**: Tasks have disjoint label spaces, and training samples arrive as a one-pass data stream without definite end/sample size. What data paradigm do you use?

Question 21 (Model-based reinforcement Learning)**.** In model-based reinforcement learning, the agent builds a model of the environment to predict state transitions and rewards. One model for how agents interact with the environment in for learning policies is SARSA: state-action-reward-state-action. We would like learn a model that given states and actions returns the next state along with a model that given the state and action predicts the reward.

- Define your datasets for learning the environment model, i.e., the transition function $P(s_{t+1}|s_t, a_t)$ and reward function $r_t(s_t, a_t)$.
- How might you use to reward model to choose the best action?

Fundementals

Question 22 (Graph equations I)**.** (I) Consider a vector function $\mathbf{x}(t) = \begin{bmatrix}\cos t & \sin t\end{bmatrix}^\top$.

a. Draw the set of points that this function passes through.

b. To build intuition, draw the velocity vector at a few points by considering the direction that the point moves in.

c. Find the derivative $\frac{d\mathbf{x}}{dt}$. Draw this vector for some point t.

Question 23 (Index to Matrix)**.** (I) Turn the following index expressions back into matrix-vector notation:

a. $\sum_{ijk} A_{ij}B_{jk}C_{ki}$

b. $b_i + \sum_j A_{ij}b_j$

c. $x_i x_j$

d. $\sum_j \delta_{ij}a_j$

Question 24 (Index notation proofs I)**.** (I) Using index notation, show that:

a. $\mathbf{x}^\top \mathrm{A}\mathbf{y} = \mathbf{y}^\top \mathrm{A}\mathbf{x}$ if A is symmetric, i.e., $\mathrm{A} = \mathrm{A}^\top$.

b. $\mathbf{x}^\top \mathbf{y} = \mathrm{Tr}(\mathbf{x}^\top \mathbf{y}) = \mathrm{Tr}(\mathbf{y}^\top \mathbf{x})$, for $\mathbf{x}, \mathbf{y} \in \mathbb{R}^D$.

c. $\mathrm{Tr}(\mathrm{ABC}) = \mathrm{Tr}(\mathrm{CAB})$.

Question 25 (Simplification of the Quadratic Form Gradient)**.** (I)
In lecture, we derived the identity:

$$\nabla_\theta(\theta^\top A\theta) = (A + A^\top)\theta.$$

In this exercise, assume that A is symmetric, i.e., $A = A^\top$, and show that the above gradient simplifies to $2A\theta$. *Hint: this is also a nice opportunity to practice index notation skills.*

Linear Regression

Question 26 (Multivariate Linear Model)**.** (I)
In our lecture we saw a linear model but we have stated it with a single dimensional output. Please generalize this to a multivariate case e.g., $\boldsymbol{y} \in \mathbb{R}^l$.

a. What is the new shape of w?

b. How does that impact the complexity of the model?

c. Imagine you find the best performing model has $W_{2,j} = 0$ where j is even. What does this imply about the relationship between inputs and outputs?

Question 27 (Expanding the Linear Regression Loss in Index Notation)**.** (I)
In our lecture, we wrote the linear regression loss in compact vector form as

$$\mathcal{L}(\theta) = ||\boldsymbol{X}\theta - \boldsymbol{y}||_2^2.$$

Using index notation, expand this expression to show that it can also be written as

$$||\boldsymbol{X}\theta - \boldsymbol{y}||_2^2 = \sum_j \left(\boldsymbol{x}^{(j)\top}\theta - y^{(j)}\right)^2.$$

Here, $\boldsymbol{x}^{(j)}$ denotes the jth row of the design matrix $\boldsymbol{X}$, corresponding to the jth training example.